

Writing and Solving Equations in Real-World Contexts*Worksheet 7A.01.07*

Name _____

Date _____

Period _____

For questions 1 through 4, define the variable in a sentence, write an equation, solve it, and say what the solution means.

1. Peyton was given a \$40 gift card to her local coffee shop. She went to the shop four times and purchased the same thing each time. She has \$7.08 left on the card. What was the cost of each purchase?

Variable.

Equation.

Solve.

Meaning.

2. Georgina was given a \$50 gift card to a music app. She bought 18 songs that all cost the same amount. She has \$18.50 left on the card. How much did each song cost?

Variable.

Equation.

Solve.

Meaning.

3. Shirley gives each of her eight grandchildren a \$15 gift card and the same amount of cash for their birthday. She spends \$280 in all. How much cash does she give each grandchild?

Variable.

Equation.

Solve.

Meaning.

4. Fatima ran 7 fewer miles this week than twice the number she ran last week. This week she ran 9 times as far as Luna, who ran 3 miles. How many miles did Fatima run last week?

Variable.

Equation.

Solve.

Meaning.

For questions 5 and 6, the coefficient is a fraction. Multiplying by the reciprocal is usually the quickest route.

5. Kiara is baking cookies for a bake sale. Each batch needs $1\frac{2}{3}$ cups of sugar. She has 12 cups, and her mother asked her to set aside $\frac{1}{3}$ cup for her coffee. How many batches can Kiara make?

Variable.

Equation.

Solve.

Meaning.

6. A craft kit uses $\frac{1}{6}$ yard of ribbon for each bracelet, plus 2.5 yards for the display board. The kit contains 9 yards of ribbon in all. How many bracelets does the kit make?

Variable.

Equation.

Solve.

Meaning.

For questions 7 and 8, decide whether your answer makes sense in the situation before you write your sentence.

7. A charter bus for a field trip costs \$340 plus \$28 for each student who rides. The budget for the trip is \$1,000. How many students can go?

Variable.

Equation.

Solve.

Meaning.

8. A city water bill charges \$0.00412 for each gallon used plus a \$22.40 service charge. Last month's bill was \$63.60. How many gallons were used?

Variable.

Equation.

Solve.

Meaning.

Questions 9 and 10 look ahead to multi-step equations, which arrive later this year. The unknown appears on both sides of the equal sign.

- 9.** Nadia charges a \$20 booking fee plus \$12 an hour to babysit. Priya charges a \$50 booking fee plus \$6 an hour. At how many hours do the two charge the same amount?

Variable.

Equation.

Solve.

Meaning.

- 10.** Solve $5(x - 3) = 3x + 11$. Check your solution by substitution.

Equation.

Solve.

Meaning.